MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2026
CHAPTER 2: SEQUENCES (PART 1)

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Goals of this section:

(1) Sequences and convergence
(2) Monotone sequences and MCT
(3) Examples

Reference books:

- Robert G. Bartle, Donald R. Sherbert: Introduction to Real Analysis (Wiley, 2011, Fourth Edition)
- Kenneth A. Ross: Elementary Analysis: The Theory of Calculus (Springer, 2013, Second Edition)

1. Sequence of numbers and convergence

1.1. **Definition.**

Definition 1.1. A sequence in $\mathbb{R}$ is a function x from $\mathbb{N}$ to $\mathbb{R}$. We shall denote the element $x(n)$ by x_n for $n \in \mathbb{N}$. We also sometimes write a sequence of the form $(x_1, x_2, \ldots)$ or (x_n). For example, if we write $(2, 4, 6, 8, \ldots)$, then this means that $x_1 = 2$, $x_2 = 4$, $x_3 = 6$, $x_4 = 8$, and so on.

Example 1.2. (a) Let $x_n = 3$ for all $n \in \mathbb{N}$. Then the sequence is the same for all $n \in \mathbb{N}$, and it takes the form $(3, 3, 3, \ldots)$.
(b) Let $r \in \mathbb{R}$. Let $x_n = r^n$. This is the so-called geometric sequence.
(c) Let $x_n = (-1)^n$. The sequence takes the form $(-1, 1, -1, 1, \ldots)$.
(d) The celebrated Fibonacci sequence (f_n) is defined as:

$$f_1 := 1, \quad f_2 := 1, \quad f_{n+1} := f_n + f_{n-1} \ (n \geq 2)$$

Exercise 1.3. (1) Let (x_n) be a sequence in $\mathbb{R}$. Explain that there is a number r in $\mathbb{R}$ such that $x_n \neq r$ for all $n \in \mathbb{N}$
(2) Describe a sequence (x_n) in $\mathbb{Q}$ such that for each number r in $\mathbb{Q}$, there is a unique $m \in \mathbb{N}$ such that $x_m = r$.

1.2. Convergence and limits.

Definition 1.4. (1) A sequence (x_n) in $\mathbb{R}$ is said to converge to x in $\mathbb{R}$, or $x \in \mathbb{R}$ is said to be the limit of a sequence (x_n) if for every $\epsilon > 0$, there exists $N(\epsilon)$ such that for any $n > N(\epsilon)$, $|x_n - x| < \epsilon$. (*We may sometimes simply write N rather than $N(\epsilon)$.*)

(2) If the limit of a sequence exists, we say that the sequence is convergent. If the limit of a sequence does not exist, then we say that the sequence is divergent.

(3) When a sequence in $\mathbb{R}$ has the limit x, we shall write

$$\lim x_n = x \quad \text{or} \quad \lim_{n\to\infty} x_n = x.$$

Exercise 1.5. Draw a real line and indicate which x_n lie in the interval $(x-\epsilon, x+\epsilon)$ in Definition 1.4.

1.3. Uniqueness of limits.

Theorem 1.6. *(Uniqueness of a limit) Each convergent sequence has a unique limit.*

Proof. Let (x_n) be a convergent sequence in $\mathbb{R}$. Let x' and x'' be limits of (x_n). Let $\epsilon = \frac{|x'-x''|}{2}$. Suppose $x' \neq x''$ and so $\epsilon > 0$ to derive a contradiction.

Step 1: Large n so that x_n is close to x' and x''

By the definition of limits, there exist integers $N'(\epsilon)$ such that for $n > N'(\epsilon)$,

$$|x_n - x'| < \epsilon$$

Similarly, there exist integers $N''(\epsilon)$ such that for $n > N''(\epsilon)$,

$$|x_n - x''| < \epsilon$$

Step 2: Get contradiction that for large n, x_n cannot be so close to both x' and x''

Now, pick $n^* > \max\{N'(\epsilon), N''(\epsilon)\}$. Then,

$$|x' - x''| = |x' - x_{n^*} + x_{n^*} - x''| \leq |x' - x_{n^*}| + |x_{n^*} - x''| < \epsilon + \epsilon = 2\epsilon.$$

However, this means $|x' - x''| < |x' - x''|$, giving a contradiction. Thus, we must have that $x' = x''$. □

Example 1.7. (a) Guess $\lim_{n\to\infty}(\frac{1}{n})$ and then use the $\epsilon - N$-definition to show that.

Sol: Given $\epsilon > 0$, we find a number $N > \frac{1}{\epsilon}$. Then, for any $n > N$,

$$|\frac{1}{n} - 0| = \frac{1}{n} < \frac{1}{N} < \epsilon.$$

By the definition of a limit, $\lim(\frac{1}{n}) = 0$.

(b) Guess $\lim(\frac{2}{n^2+2})$ and then use $\epsilon - N$ definition to show that.
Sol: Guess

$$\lim_{n\to\infty} \frac{2}{n^2+2}$$

and then use the ϵ-N definition to prove it. *Sol.* We claim that

$$\lim_{n\to\infty} \frac{2}{n^2+2} = 0.$$

Given $\epsilon > 0$, we want

$$\frac{2}{N^2+2} < \epsilon.$$

Equivalently,

$$N^2 + 2 > \frac{2}{\epsilon}.$$

If $\epsilon \geq 1$, choose $N = 1$. Then, for any $n > 1$,

$$\frac{2}{n^2+2} < 1 \leq \epsilon.$$

If $\epsilon < 1$, choose

$$N > \sqrt{\frac{2}{\epsilon} - 2}.$$

Then

$$N^2 > \frac{2}{\epsilon} - 2,$$

and hence

$$\frac{2}{N^2+2} < \epsilon.$$

Therefore, for any $n > N$,

$$\left|\frac{2}{n^2+2} - 0\right| = \frac{2}{n^2+2} < \frac{2}{N^2+2} < \epsilon.$$

Hence,

$$\lim_{n\to\infty} \frac{2}{n^2+2} = 0.$$

Exercise 1.8. (Example of non-convergent sequence) Let $x_n = (-1)^n$. Show that (x_n) is divergent by the $\epsilon - N$ definition of a limit.

Exercise 1.9. (Tail of sequences) Let (x_n) be a sequence in $\mathbb{R}$. Let $m \in \mathbb{N}$. Define a new sequence $y_n = x_{m+n}$. Use the $\epsilon - N$-argument to show that (x_n) is convergent if and only if (y_n) is also convergent. Moreover, if one of the limits exists, then $\lim y_n = \lim x_n$.

In the statements below, such as Lemma 2.4 and Theorem 3.6, we often assume that an inequality holds for all $n \in \mathbb{N}$. Using tail sequences, the same conclusions remain valid if the inequality holds only eventually; that is, if there exists $m \in \mathbb{N}$ such that it holds for all $n > m$.

2. Boundedness of a convergent sequence

2.1. Convergent ⇒ Bounded.

Definition 2.1. A sequence (x_n) in $\mathbb{R}$ is said to be bounded if the set $\{x_n : n \in \mathbb{N}\}$ is bounded i.e. there exist numbers $L, M \in \mathbb{R}$ such that $L < x_n < M$ for all $n \in \mathbb{N}$.

Theorem 2.2. *Any convergent sequence in $\mathbb{R}$ is bounded.*

Proof. Let (x_n) be a convergent sequence, and let $x = \lim x_n$. Pick $\epsilon = 1$. Then there exists an integer $N \in \mathbb{N}$ such that for any $n > N$, $|x_n - x| < 1$ and so $x_n < x + 1$. Let $M = \max\{x + 1, x_1, \ldots, x_N\}$. Then $x_n \leq M$ for any $n \in \mathbb{N}$, and hence $\{x_n : n \in \mathbb{N}\}$ is bounded above.

The argument for boundedness below is similar. □

Example 2.3. Let $x_n = \sqrt{n + \frac{1}{n}}$. The sequence (x_n) is not bounded and so is divergent.

2.2. Bounding a convergent sequence.

Lemma 2.4. *Let (x_n) be a convergent sequence in $\mathbb{R}$. If there exist $L, M \in \mathbb{R}$ such that $L \leq x_n \leq M$ for all $n \in \mathbb{N}$, then $L \leq \lim x_n \leq M$.*

Proof. Let $x = \lim x_n$. Suppose $x > M$. Let $\epsilon > 0$ such that $x - \epsilon > M$ (why such ϵ exists?). By definition of limit, there exists $N \in \mathbb{N}$ such that for any $n > N$,

$$|x_n - x| < \epsilon$$

and so $x - \epsilon < x_n < x + \epsilon$.

The last inequality implies there exists $n^* \in \mathbb{N}$ such that $x_{n^*} > x - \epsilon > M$. This gives a contradiction. Hence $x \leq M$. Proving $L \leq x$ is similar. □

The following reformulation is useful later:

Corollary 2.5. *Let (x_n) be a convergent sequence in $[a, b]$. Then* $\lim\ x_n \in [a, b]$.

Exercise 2.6.
(a) Does the converse of Theorem 2.2 hold?
(b) Could we replace the non-strict inequalities in Lemma 2.4 by strict inequalities?

3. Some basic methods of showing convergence

3.1. Construct convergent sequences by basic operations.

We shall explain how to use some simple operations such as additions and multiplications to obtain convergence from known ones.

Theorem 3.1. *Let (x_n) and (y_n) be two convergent sequences in $\mathbb{R}$. Let $x = \lim x_n$ and let $y = \lim y_n$.*

(a) *For $c \in \mathbb{R}$, (cx_n) converges to cx.*
(b) *$(x_n + y_n)$ converges to $x + y$.*
(c) *$(x_n y_n)$ converges to xy.*
(d) *$(-x_n)$ converges to $-x$.*
(e) *$x_n - y_n$ converges to $x - y$.*
(f) *If $x_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim x_n \neq 0$, then $\lim \frac{1}{x_n} = \frac{1}{x}$.*

Proof. For any $\epsilon > 0$, there exist integers $N_x(\epsilon)$ and $N_y(\epsilon)$ such that

$$n > N_x(\epsilon) \Rightarrow |x_n - x| < \epsilon, \qquad n > N_y(\epsilon) \Rightarrow |y_n - y| < \epsilon.$$

Let

$$N(\epsilon) = \max\{N_x(\epsilon), N_y(\epsilon)\}.$$

Then for all $n > N(\epsilon)$, both inequalities hold.
Prove (a). Let $\epsilon > 0$. If $c = 0$, a constant sequence is always convergent and (cx_n) converges to 0. Suppose $c \neq 0$. Then, for $n > N(\frac{\epsilon}{|c|})$,

$$|cx_n - cx| = |c||x_n - x| < |c|\frac{\epsilon}{|c|} = \epsilon.$$

Prove (b). Let $\epsilon > 0$. Then, for $n > N(\frac{\epsilon}{2})$,

$$|(x_n + y_n) - (x + y)| = |(x_n - x) + (y_n - y)| \leq |x_n - x| + |y_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Prove (c). *Step 1: Bound sequences.* Let $\epsilon > 0$. Then, by Theorem 2.2, (x_n) and (y_n) are bounded, and so there exists a number M in $\mathbb{R}$ such that $|x_n| < M$ and $|y_n| < M$.
Step 2: Bound limits By Lemma 2.4, $|y| \leq M$ (we also have $|x| \leq M$, but we do not need this).
Step 3: Form inequalities Then, for $n > N(\frac{\epsilon}{2M})$,

$$|x_n y_n - xy| = |x_n y_n - x_n y + x_n y - xy| \leq |x_n y_n - x_n y| + |x_n y - xy| < M|y_n - y| + M|x_n - x| = \epsilon.$$

Prove (d). Let $\epsilon > 0$. Then, for any $n > N(\epsilon)$,

$$|(-x_n) - (-x)| = |-x_n + x| = |x_n - x| < \epsilon.$$

Prove (e). This follows from (b) and (d).

Prove (f).(Sketch) *Step 1:* First show that there exists a real number $M > 0$ such that $|x_n| > M$. (Hint: Similar argument to the proof of Theorem 2.2)

Step 2: Show that $|x| \geq M$ by Theorem 2.4.

Step 3: Let $\epsilon > 0$. Then, for any $n > N(M^2\epsilon)$,

$$\left|\frac{1}{x_n} - \frac{1}{x}\right| = \frac{|x - x_n|}{|x_n||x|} \leq \frac{1}{M^2}|x - x_n| < \epsilon.$$

□

Example 3.2. Let $x_n = 2\left(1 - \frac{1}{n}\right)$ for all $n \in \mathbb{N}$. Find $\lim x_n$.
Sol: It follows from Theorem 3.1 that

$$\lim x_n = 2 - 2\lim \frac{1}{n} = 2 - 2(0) = 2.$$

Exercise 3.3. Let k be a positive integer. Let (x_n) be a sequence converging to x. Show that $\lim x_n^k = x^k$.

Exercise 3.4. Let $x_n = \frac{n^2+2n+3}{3n^2+4n+5}$. Find $\lim\ x_n$.

3.2. **Squeeze theorem.**

Exercise 3.5. Let $(x_n), (y_n)$ and (z_n) be convergent sequences in $\mathbb{R}$. Suppose

$$x_n \leq y_n \leq z_n$$

for all $n \in \mathbb{N}$. Then

$$\lim\ x_n \leq \lim\ y_n \leq \lim\ z_n.$$

Theorem 3.6. *(Squeeze theorem) Let $(x_n), (y_n), (z_n)$ be three sequences in $\mathbb{R}$. Suppose*

$$x_n \leq y_n \leq z_n$$

for all $n \in \mathbb{N}$. Suppose x_n and z_n are both convergent and $\lim x_n = \lim z_n$. Then (y_n) is also convergent and $\lim y_n = \lim x_n$.

Proof. Let $w = \lim\ x_n$. Given $\epsilon > 0$, there exists an integer N such that for all $n > N$,

$$|x_n - w| < \epsilon, \quad |z_n - w| < \epsilon$$

and so $-\epsilon < x_n - w < \epsilon, \quad -\epsilon < z_n - w < \epsilon$.

On the other hand,

$$x_n - w \leq y_n - w \leq z_n - w$$

and so $-\epsilon < y_n - w < \epsilon$. Thus, $|y_n - w| < \epsilon$. □

Example 3.7. See the book of Bartle-Sherbert (f) Page 67.

Example 3.8. We show that

$$\lim_{n\to\infty} (\sqrt{n+1} - \sqrt{n}) = 0.$$

For every $n \in \mathbb{N}$,

$$\sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}.$$

Moreover,

$$0 < \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}}.$$

Since

$$\lim_{n\to\infty} \frac{1}{2\sqrt{n}} = 0,$$

the squeeze theorem implies that

$$\lim_{n\to\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0.$$

Therefore,

$$\lim_{n\to\infty} (\sqrt{n+1} - \sqrt{n}) = 0.$$

Exercise 3.9. Let (x_n) be a sequence in $\mathbb{R}$. Let $x \in \mathbb{R}$. Let (a_n) be a sequence of positive numbers such that $\lim a_n = 0$. Suppose, for all $n \in \mathbb{N}$,

$$|x_n - x| \leq a_n.$$

Show that $\lim\ x_n = x$.

4. Monotone Convergence Theorem

4.1. **Monotone sequences.**

Definition 4.1. A sequence in $\mathbb{R}$ is said to be increasing if it satisfies the inequality:

$$x_1 \leq x_2 \leq x_3 \leq \dots$$

A sequence in $\mathbb{R}$ is said to be decreasing if it satisfies the inequality:

$$x_1 \geq x_2 \geq x_3 \geq \dots$$

A sequence is said to be monotone if it is either increasing or decreasing.

Example 4.2.
(1) The sequence (2^n) is increasing.
(2) The sequence $(n+2)$ is increasing.
(3) The sequence $(\frac{1}{2^n})$ is decreasing.
(4) The sequence $(\frac{1}{n^2+2})$ is decreasing

4.2. **A bounded criterion.** We have the following criteria for the convergence of a monotone sequence:

Theorem 4.3. *(Monotone Convergence Theorem) A monotone sequence is convergent if and only if it is bounded.*

Proof. We shall consider the sequence to be increasing. The situation for a decreasing sequence is similar.
If direction:
Suppose the sequence (x_n) is bounded. Let

$$s = \sup\{x_n : n \in \mathbb{N}\},$$

which exists by the completeness property. Then for any given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $s - \epsilon < x_N$. Since the sequence is increasing, for any $n > N$,

$$x_n \geq x_N > s - \epsilon.$$

Thus, $s - x_n < \epsilon$. Since $x_n \leq s$ for all n, we have $0 \leq s - x_n < \epsilon$. Thus $|s - x_n| < \epsilon$. By definition, s is the limit of (x_n).

Only if direction:
This follows from Theorem 2.2. □

The proof of Theorem 4.3 gives that:

Corollary 4.4. (1) *If (x_n) is a bounded increasing sequence, then*

$$\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}.$$

(2) *If (x_n) is a bounded decreasing sequence, then*

$$\lim(x_n) = \inf\{x_n : n \in \mathbb{N}\}.$$

Remark: The above corollary does not hold if one removes the condition of monotone. For example, take $x_1 = 3$, for $n \geq 2$, $x_n = 2 - \frac{1}{n}$. Then the sequence is convergent and $\lim\ x_n = 2$, but $\sup\{x_n : n \in \mathbb{N}\} = 3$.

4.3. **Examples and applications.**

Example 4.5. Let $1 < x_1 < 2$. Let $x_{n+1} = 1 + \sqrt{x_n - 1}$. Show that $\lim x_n$ exists in $\mathbb{R}$ and find it.
Proof: We shall show that $x_n < x_{n+1}$ for all $n \in \mathbb{N}$ and $1 < x_n < 2$. We shall proceed by mathematical induction.

For x_1, it follows from the given condition. Suppose the statement is true for certain $m \in \mathbb{N}$. Since $x_m - 1 < 1$, $0 < \sqrt{x_m - 1} < 1$ and so $0 < x_m - 1 < \sqrt{x_m - 1}$. Thus, $x_m - 1 < x_{m+1} - 1$ and so $x_m < x_{m+1}$. This then also implies that $1 < x_{m+1}$

since $1 < x_m$. We also have $x_{m+1} < 2$ since $x_m - 1 < 1 \Rightarrow \sqrt{x_m - 1} < 1 \Rightarrow 1 + \sqrt{x_m - 1} < 2$.

Since (x_n) is also bounded, the monotone convergence theorem also implies that $\lim x_n$ exists. Let $L = \lim x_n$. Then

$$\lim(x_{n+1} - 1)^2 = \lim(x_n - 1)$$

and so

$$(\lim\ x_{n+1} - 1)^2 = \lim\ x_n - 1.$$

Hence, $(L-1)^2 = (L-1)$ and so $(L-1)(L-2) = 0$. Then, $L = 1$ or 2. Since $1 < x_1 \leq L$, we can only have $L = 2$.

Exercise 4.6. Define $x_1 = \sqrt{3}$. Define $x_n^2 = 3x_{n-1}$ and $x_n > 0$ for $n \geq 2$. Show that $\lim x_n$ exists and find it.

5. Some useful limits

5.1. Geometric Sequence.

Theorem 5.1. (1) *If $0 < r < 1$, then $\lim_{n\to\infty} r^n = 0$.*

(2) *If $r > 1$, then (r^n) is divergent.*

Proof. (1) Let $x_n = r^n$. Since $x_{n+1} = r(x_n) < x_n$, the sequence (x_n) is decreasing and so monotone. Since (x_n) is bounded, the MCT implies that the sequence is convergent. Let $x_n = r^n$. Then $x_{n+1} = rx_n$ and so $\lim(x_{n+1}) = \lim(rx_n)$. Then, $L = rL$ and so $L = 0$.

(2) Let $x_n = r^n$. If $\lim_{n\to\infty} x_n$ exists, let $L = \lim_{n\to\infty} x_n$. Since $x_{n+1} = rx_n$, $\lim_{n\to\infty} x_{n+1} = r\lim_{n\to\infty} x_n$ and so $L = rL$. Thus, $L = 0$. But $x_n > 1$ for all n and $L \geq 1$. This gives a contradiction.

□

5.2. $n^{1/n}$.

Theorem 5.2.

$$\lim n^{1/n} = 1.$$

Proof. Let $x_n = n^{1/n} - 1$. For $n > 1$, $x_n > 0$. We now consider

$$n = (x_n + 1)^n$$

and so

$$n = 1 + nx_n + \frac{n(n-1)}{2}x_n^2 + \ldots > 1 + \frac{n(n-1)}{2}x_n^2.$$

Hence, $(n-1) > \frac{n(n-1)}{2}x_n^2$ and so

$$\frac{2}{n} > x_n^2.$$

This gives $\sqrt{\frac{2}{n}} > x_n > 0$. Since $\lim \sqrt{\frac{2}{n}} = 0$, by Squeeze Theorem,

$$\lim x_n = 0.$$

Hence, $\lim n^{1/n} = \lim\ (x_n + 1) = 1$. □

Exercise 5.3. Find the limits of the following sequences:

(1) $\lim\ n^{1/n^3}$
(2) $\lim\ (n!)^{1/n}$
(3) For $a > 0$, $\lim\ a^{1/n}$.

5.3. **Euler number.**

Theorem 5.4. *The sequence* $\left(\left(1 + \frac{1}{n}\right)^n\right)$ *converges. The limit will be denoted by the number* e.

Proof. Define $x_n = \left(1 + \frac{1}{n}\right)^n$. Then,

$$\begin{aligned} x_n &= \left(1 + \frac{1}{n}\right)^n \\ &= 1 + \frac{n}{1} \cdot \frac{1}{n} + \frac{n(n-1)}{2} \cdot \frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!} \cdot \frac{1}{n^3} + \ldots + \frac{n(n-1)\ldots 1}{n!} \cdot \frac{1}{n^n} \\ &= 1 + 1 + \frac{1}{2}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \ldots + \frac{1}{n!} \cdot \left(1 - \frac{1}{n}\right) \ldots \left(1 - \frac{n-1}{n}\right) \end{aligned}$$

Increasing: Note that x_{n+1} has similar expressions and the middle terms look like:

$$\frac{1}{k!}\left(1 - \frac{1}{n+1}\right) \ldots \left(1 - \frac{k-1}{n+1}\right)$$

which is greater than the corresponding term

$$\frac{1}{k!}\left(1 - \frac{1}{n}\right) \ldots \left(1 - \frac{k-1}{n}\right)$$

in x_n. Moreover, x_{n+1} has one more term than x_n, and so $x_{n+1} > x_n$. This shows that (x_n) is increasing.

Bounded above: Note that

$$\frac{1}{k!}\left(1 - \frac{1}{n}\right) \ldots \left(1 - \frac{k-1}{n}\right) \leq \frac{1}{k!} \leq \frac{1}{2^{k-1}}$$

Thus,

$$x_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \ldots < 3.$$

Conclusion:
By the monotone convergence theorem, the sequence (x_n) converges. □

Exercise 5.5. Find the limit of the following sequences:

(1) $\left(1+\frac{1}{n+1}\right)^n$;
(2) $\left(1+\frac{1}{n}\right)^{2n}$.

5.4. $n\sin\left(\frac{1}{n}\right)$.

Exercise 5.6. Find, using trigonometry, that

$$\lim_{n\to\infty} n\sin\left(\frac{1}{n}\right)$$